MaterialNeutral:
Revealing Active Sites in Ammonia Decomposition over Ru supported on hydrothermally-derived and post-HTC pyrolyzed Carbon

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Introduction

Ruthenium (Ru) has emerged as one of the most effective catalysts for ammonia decomposition, presenting a promising pathway for hydrogen production in a sustainable energy economy [1]. Among the carbon-based supports, hydrothermally-derived carbon has gained significant attention due to its tunable surface properties and high porosity [2]. As an environmentally friendly and resource-efficient approach, hydrothermal carbonization (HTC) enables the transformation of biomass into carbon-rich materials under mild conditions. [3]. Additional thermal treatment can further optimize its support properties [4]. This study aims to reveal the active sites involved in ammonia decomposition over Ru supported on hydrothermally-derived and post-HTC pyrolyzed carbon, focusing on both structural and catalytic stability to design robust catalysts.

Experimental

The carbon materials were synthesized via hydrothermal carbonization (HTC) at 180 °C for 16 h. Further, a pyrolysis step at temperatures between 600 °C and 1000 °C was carried out to tune the graphitization degree, hence the thermal stability. While using Ru(III) nitrosyl nitrate as a precursor, Ru nanoparticles were loaded onto the carbon support by wet impregnation. By this procedure, four catalysts, i.e. Ru/HC, Ru/HC-600, Ru/HC-800 and Ru/HC-1000 were produced with the number indicating the respective pyrolysis temperature.

Besides comprehensive characterization of the produced materials, the catalytic activity of the catalysts was screened in a fixed-bed flow reactor measuring the NH_3 conversion. The experiments were conducted at ambient pressure and temperatures

ranging from 300 °C to 600 °C, with space velocities from 7,500 NmL h^{-1} g^{-1} to 30,000 NmL h^{-1} g^{-1}.

Results and Conclusion

The HTC yield was assessed to be ~50 wt.-% (± 2 wt.-%) with the carbon content rising to 90 wt.-% for the sample subjected to the highest pyrolysis temperature, as illustrated in Fig. 1 a).The increase in disorder, indicated by the rising D band in Fig. 1 b), results from the loss of functional groups due to dehydration, decarboxylation, and the rearrangement of carbon atoms. Notably, NH_3 conversion improved with an increase in the pyrolysis temperature of the support which could be attributed to a more uniform dispersion of Ru. In Fig. d), TEM images reveal that Ru forms larger agglomerates (7 nm) when directly supported on hydrochar. As pyrolysis temperature increases, the dispersion becomes more homogenous, with Ru nanoparticles averaging 3-4 nm in size. Reproducibility of the wet impregnation procedure was proven by a catalyst replicated from the same support batch, as shown in Fig 1 e). However, when preparing HTC catalysts using the same procedure across different raw material batches, the activity varied, despite the Ru content remaining constant. Investigation of the underlying effects will be a key focus of this work.

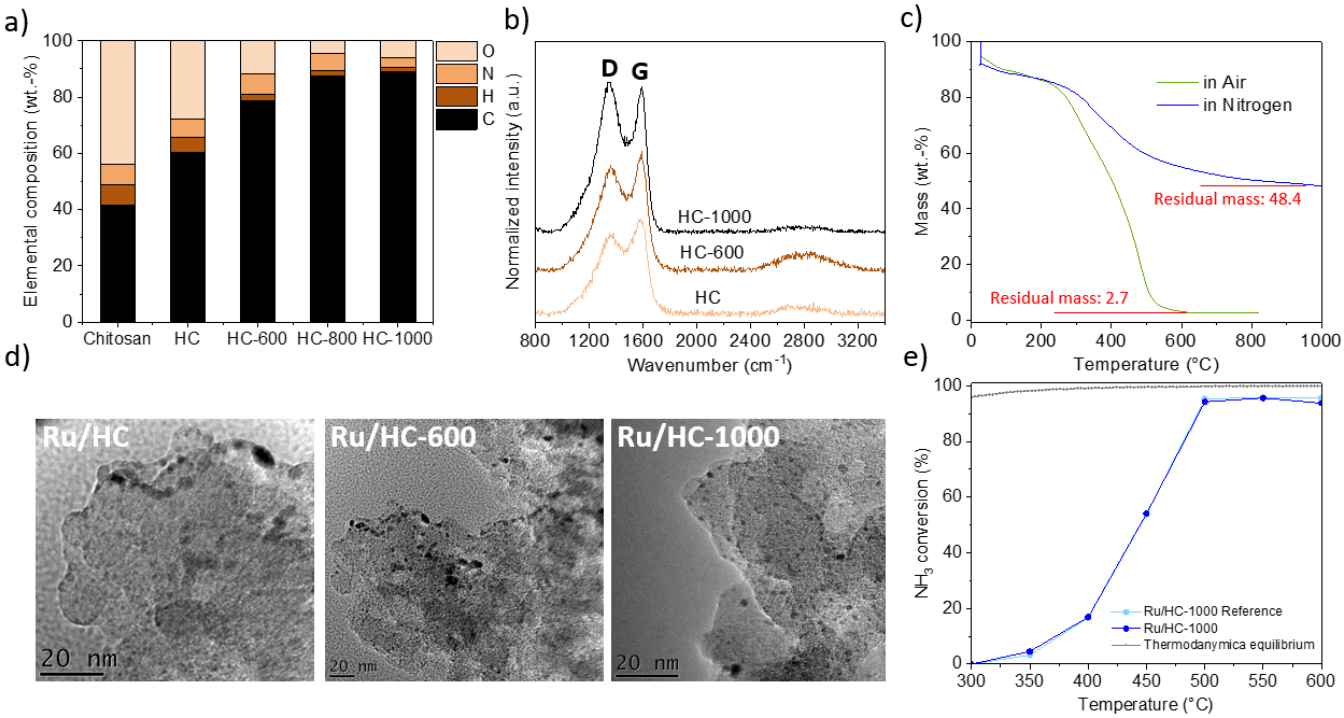

Figure 1. Support and catalyst properties: a) elemental composition of the supports, b) raman spectra of the supports, c) Air- and N_2-TGA of the hydrochar (HC), d) TEM bright-field images, e) catalytic activity of Ru/HC-1000 with a SV of 15.000 NmL h^{-1} g^{-1}.

Significance

The findings of this study advance a sustainable and eco-friendly ammonia decomposition process by unlocking lower reaction temperatures compared to the Ni-based systems. Furthermore, bio-based materials will be used for catalyst production with the potential of being carbon-negative over the lifetime of the catalyst with further potential of easy recyclability.

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